Tahsin Touheed Ivan

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Education

University of Dhaka — Dhaka, Bangladesh
B.Sc. in Electrical and Electronic Engineering (4th year) — *Sept 2022 – Dec 2026 (expected)*
CGPA: 3.73 / 4.00

Notre Dame College — Dhaka, Bangladesh
Higher Secondary Certificate (Science) — *Aug 2019 – July 2021*
GPA: 5.00 / 5.00

Ideal School & College — Dhaka, Bangladesh
Secondary School Certificate (Science) — *Jan 2009 – July 2019*
GPA: 5.00 / 5.00

Research

Materials Science Research Laboratory (MSRL DU) — Jan 2026 – Present
Undergraduate Researcher | Supervisor: Imtiaz Ahmed
Interests: Nanomaterials Synthesis & Characterization, Machine Learning for Materials Discovery, Energy Storage Materials, Electronic & Photonic Device Engineering.

Current Work — *Machine-Learning-Guided Design of Porous Carbon & Metal-oxide based Electrodes for Supercapacitors*

- Developing a structured literature dataset of porous carbon electrodes and applying machine learning to uncover quantitative relationships between structural properties — BET surface area, pore size distribution, nitrogen/oxygen heteroatom content — and specific capacitance, aiming to identify high-performance material configurations through data-driven screening.
- Experimentally validating model-shortlisted configurations across the full pipeline: precursor carbonization and chemical activation, structural characterization (BET, Raman, XPS), and electrochemical testing (CV, GCD, EIS) — closing the loop between computational prediction and physical material behavior.

Leadership

IEEE SIGHT Student Branch, DU — Feb 2025 – Present

Chair — *April 2026 – Present*
- Supervise development of technology-based solutions targeting real-world societal challenges, and drive collaborations with IEEE chapters to expand technical and community impact.

Program Coordinator — *Feb 2025 – March 2026*
- Coordinated and organized the *Machine Learning Fundamentals* workshop for undergraduate students — managed logistics, content planning, and participant engagement.
- Led a hands-on session in the *IEEE Day Outreach Programme on Microcontrollers and Their Utilization* — jointly organized by IEEE SB University of Dhaka, IEEE EDS SBC DU, and IEEE WIE AG SB DU at Nawab Faizunnesa Govt Girls High School, Cumilla — for 50+ high school students.

IEEE Photonics Society Student Branch, DU — Feb 2025 – March 2026
Event Coordinator
- Organized technical workshops, seminars, and community outreach programs.

Optics and Photonics Club — Jan 2024 – April 2025
Vice President
- Worked on projects, seminars, and events hand-in-hand with OPTICA.
- Delivered talks and project showcases (LFR, Holography, etc.) at the outreach program *Envisioning Harmony Between Human and Nature*, St. Joseph Higher Secondary School, Dhaka — in collaboration with the Optica Bangladesh Student Chapter, University of Dhaka, and Scintilla Science Club.

Projects

Project: Drone (Funded by FabLab DU) | *Python, Arduino, ESP32* 2024

- Co-developed FabLab DU's first departmental drone (ESP32 quadcopter UAV, UoD EEE) for ∼USD 66 — integrated an F450 Quad-X frame, A2212 1000KV BLDC motors, 30A ESCs, MPU6050 IMU, and FlySky FS-i6 RC system on an ESP32 DevKit V1 running ESP-FC/Betaflight firmware; achieved 2.8:1 TWR, ±5° hover stability, ∼15 m altitude, and ∼13 min flight time. Architected for extensibility with GPS (Neo-6M), barometric altimeter (BMP280), FPV camera, LiDAR obstacle avoidance, and DSHOT ESC upgrade — without a flight-controller hardware change.
- Gained end-to-end embedded flight-systems experience: configured ESP32–ESC PWM communication (1000–2000 μs) and PPM/SBUS RC integration; implemented and hand-tuned cascaded PID control (rate + angle loops) across roll, pitch, and yaw; diagnosed and resolved ESC desync, IMU yaw drift, and vibration instability; explored ESP32's dual-core architecture and onboard Wi-Fi as a basis for telemetry, autonomous navigation, and sensor fusion — validating commodity microcontrollers as low-cost alternatives to STM32-based commercial flight controllers.

ESP32 Retro Gaming Console | *ESP32* 2026

- Built a handheld retro gaming console prototype (ESP32, 0.96" SSD1306 OLED, analog joystick, passive buzzer) running three original arcade games (Car, Snake, Pong) with a scrollable menu, idle sleep animation, and per-game high scores saved to flash.
- Focused on ESP32 real-time firmware design — millis()-based finite-state machines, ADC noise filtering with hysteresis, and debounce logic — alongside peripheral integration: I^2C display rendering with U8g2, LEDC PWM audio, and NVS for non-volatile storage across deep-sleep wake cycles.

Pocket Reader — ESP32 Handheld E-Book Reader | *ESP32* 2026

- Built a self-contained handheld e-book reader prototype (ESP32, 0.96" SSD1306 OLED, analog joystick) serving a mobile-friendly web portal directly from the microcontroller for book uploads and Wi-Fi management — no cloud, no external server. Designed for extensibility toward an e-ink display and a distraction-free writing mode.
- Focused on ESP32 networking — Wi-Fi station and access-point modes, mDNS, and an on-device REST HTTP server — alongside NVS, LittleFS, and deep-sleep power cycles; reserved BLE hardware to preserve a future phone-sync feature without requiring shared Wi-Fi.

Laser Harp | *Arduino* 2023

- Sophomore academic project — played a two-octave span of musical notes using a laser-and-LED array.

Competencies

Programming: Python, C/C++, x86 Assembly, Verilog (RTL design), HTML, LaTeX
Software & Tools: PSPICE, Proteus, MATLAB, Simulink, AutoCAD, EasyEDA
Languages: English (IELTS 7.0), Bangla (native)
Soft Skills: Public Speaking, Team Leadership, Mentoring, Event Management, Media Coordination & Direction

References

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Additional references available upon request.